

Midterm Exam – Date: ____/____/____

Student name/UIN _____

Instructions

- Questions will be distributed in class.
- Keep calm and do it! Questions are straight-forward.
- It should not take more than 90 minutes to complete the exam.
- Please submit your completed responses before the end of the class!
- LATE EXAM WILL NOT BE CONSIDERED. If you are unable to take the exam because of illness, accident, or any other circumstances beyond your control, notify me. A university official excuse will be required. A make-up test will be scheduled asap.
- 1 point per question, 10 points total.
 - o Answer questions 1 through 7 to the best of your ability, based on your knowledge and what you have learned so far in the course.
 - o Select any three questions from 8 through 15; answer only those you have chosen.
- During the exam time I will not answer questions related to the content. Save questions for later.
- Do not contemplate cheating! You are on your honor that you work on the exam exclusively by yourself.
- Do not use notes, textbook, etc. Do not access internet as well.
- DO NOT SHARE THE EXAM QUESTIONS AND ANSWERS.

Questions

Q1) Can you explain what a Geographic Information System (GIS) is and discuss how spatial data engineering principles are applied to design, build, and optimize data pipelines that support large-scale systems, infrastructure planning, logistics networks, and decision-making? Consider aspects such as spatial data modeling, geospatial ETL processes, scalability, real-time data integration, and how spatial analytics can inform optimization and systems-level decision support.

Q2) What spatial data models are used in modern geospatial data engineering systems, and how do their structural and computational characteristics influence system design, storage strategies, and analytical performance? In particular compare raster and vector models in terms of data representation, scalability, distributed processing, and suitability for large-scale industrial, infrastructure, environmental, or logistics applications.

Q3) What is the difference between a datum and a projection in GIS? Please provide appropriate examples of each and explain why they are important. In your response, discuss the role of spatial scale (local, regional, and global) in selecting an appropriate projected coordinate reference system (CRS), and explain why choosing a suitable projected CRS is critical when performing spatial operations.

Q4) How have you designed and implemented geospatial data engineering workflows to support large-scale analytics? How do these workflows integrate with industrial and systems engineering decision processes?

Q5) When aggregating granular hazard data such as the FEMA flood zones into 100-year and 500-year categories: What information is lost? Under what analytical, operational, or optimization contexts is such aggregation justified? How might collapsing categories obscure nonlinear risk gradients, uncertainty bounds, or edge-case vulnerabilities that are critical for resilient system design?

Q6) Describe a data dictionary or metadata documentation you have developed. What information did you include. How did it support data management and analysis?

Q7) What is spatial autocorrelation in the context of spatial data engineering and statistical modeling, and how does spatial dependence influence the design, validation, and interpretation of large-scale analytical systems? Can spatial autocorrelation affect model assumptions (e.g., independence) and the design of scalable geospatial analytics pipelines?

Q8) What are the limitations of using GIS? Under what condition might the technology hinder problem solving rather than help?

Q9) How do you typically handle missing or inconsistent values during the data pre-processing stage?

Q10) What are the key components of a GIS database? How do you ensure data accuracy and integrity?

Q11) Identify and briefly describe what are the parts of a map.

Q12) Identify and describe two cartographic generalizations that you are aware in GIS.

Q13) Identify and describe two spatial operations used in GIS.

Q14) Can you describe a time when you used spatial analysis to support decisions?

Q15) How do you handle large datasets in GIS to ensure efficient processing and visualization?

Answers:

